

Functional Tests

S7-1200 Intersection Control

Basic functional tests were performed on the physical Siemens S7-1200 setup and WinCC HMI to check the main operating modes, transitions, counters and fault handling.

No.	Test	Expected result	Result
1	AUTO sequence	Complete traffic-light sequence operates correctly.	PASS
2	Vehicle request (S3 / HMI)	A request starts the programmed crossing sequence.	PASS
3	START	Physical and HMI START controls work correctly.	PASS
4	STOP	Physical and HMI STOP controls bring the system to the intended safe state.	PASS
5	Night mode	Both yellow lights flash and the system enters/leaves night mode correctly.	PASS
6	AUTO -> MANUAL	The controller reaches manual mode through a valid transition.	PASS
7	Manual direction A	Direction A opens, closes and can be cancelled correctly.	PASS
8	Manual direction B	Direction B opens, closes and can be cancelled correctly.	PASS
9	MANUAL -> AUTO	Return to AUTO works correctly from the tested manual states.	PASS
10	MANUAL -> NIGHT	The system changes from manual to night mode correctly.	PASS
11	Fault handling	Invalid green-light condition activates fault state 99 and fault indication.	PASS
12	Fault reset	Fault can be reset using the physical button and HMI.	PASS
13	Counters	Vehicle, request and cycle counters increment correctly.	PASS
14	HMI	Controls, lamps and current-step indication correspond to PLC operation.	PASS
15	Safety check	Both directions are never green at the same time during normal operation.	PASS

Result: All tested functions operated as expected.

This document is a simple functional check of an educational PLC project and is not a formal safety certification.